

DNO Surrogate: Diagnostics and Approaches Under Investigation

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Abstract

This document is the single working reference for the effort to improve long-horizon rollout accuracy of a neural surrogate for the 1D Zakharov water-wave Dirichlet–Neumann operator. It covers: (i) what is wrong with current rollouts, established by forensic analysis on the existing FNO and SpectralDNO checkpoints, plus a physical/numerical sensitivity sweep that rules out integrator and amplitude artifacts; (ii) why the training data is not the bottleneck; (iii) the three approaches currently in flight — a one-step pushforward training loss, a relative-Hamiltonian conservation loss, and a Craig–Sulem-shaped DNO architecture; and (iv) three candidates queued for the next iteration.

Current state (5 June 2026). The diagnostic case is closed: the rollout discrepancy is model-driven, not a data, integrator-resolution, or amplitude artifact. The training-loss approaches (pushforward, H-cons) reduce error on soliton-like and modal-instability regimes but regress on broadband random-sea ICs; the gap is qualitatively similar in both, suggesting the SpectralDNO architecture is the binding constraint. The Craig–Sulem-shaped architecture is currently training from scratch as the load-bearing test of that hypothesis; an H-cons SpectralDNO is retraining in parallel for an apples-to-apples comparison.

1 Problem statement

The 1D Zakharov system couples surface elevation $\eta(x, t)$ and surface velocity potential $\xi(x, t)$ on a periodic domain of length L via the Dirichlet–Neumann operator $\mathcal{G}(\eta; h)$:

$$\partial_t \eta = \mathcal{G}(\eta; h) \xi, \tag{1}$$

$$\partial_t \xi = -g \eta - \frac{1}{2} \xi_x^2 + \frac{1}{2} \frac{(\mathcal{G}(\eta) \xi + \eta_x \xi_x)^2}{1 + \eta_x^2}. \tag{2}$$

A neural surrogate $\mathcal{G}_\theta$ is trained to predict $\mathcal{G}(\eta; h) \xi$ from snapshots of (η, ξ, h) and then iterated by an explicit time integrator at evaluation time. Pointwise prediction is accurate; long-horizon rollouts are not. The shared baseline objective is the data-distribution Sobolev- H^k loss

$$\mathcal{L}_0 = \|\mathcal{G}_\theta(\eta; h) \xi - \mathcal{G}_{\text{true}}(\eta; h) \xi\|_{H^k}^2,$$

with $k = 1$ in the current configuration. Two surrogates have been trained on this baseline: a 6-block FNO1d (width 128, modes 64, FiLM depth conditioning, ~ 13.1 M parameters) and a 6-block SpectralDNO (width 128, modes 64, latent 64, ~ 17.0 M parameters). The SpectralDNO outputs a residual of the form $C(\eta, h)^\top F(\eta, h) C(\eta, h) \xi$ added to a closed-form linear-DNO baseline, and is therefore linear in ξ by construction.

2 Diagnostic findings

2.1 Failure mode: coherent in-band error

Across the worst finite-error initial conditions in the 10-regime eval suite (Tanaka, Benjamin–Feir, Stokes, random sea, linear), $\text{rel-}L^2$ error grows linearly or with slight super-linearity in time (r_{lin}^2 between 0.53 and 0.93). The error spectrum is dominated by the same wavenumber band as the truth spectrum (per-IC cosine similarity 0.57–1.00 for FNO); the high- k “Gibbs” fraction is below 0.18 in every worst case. NaN blow-ups in `bf_g1` and `bf_modal` occur at $t = 81$ – 189 of $T_{\text{max}} = 200$ while $|\eta|_{\text{max}}$ stays small (0.04–0.12). This is *numerical divergence after extended coherent drift, not a steepening cascade or a high- k artifact*.

A specific smoking gun on the linear regime: FNO at $t = 2$ has 99% of its error in sideband modes $k = 10$ – 50 , while the truth has all its energy at $k = 6$. The FNO is hallucinating nonlinear sidebands in a regime where the dynamics are strictly per-mode dispersive — a structural consequence of mixing ξ into the network’s nonlinear features. SpectralDNO, which is linear in ξ by construction, cannot produce this kind of error and dominates FNO on the linear regime by -71% in median $\text{rel-}L^2$.

2.2 Physics descriptors that predict failure

Within each regime, the strongest Spearman correlations with final-time $\text{rel-}L^2$ are:

- Tanaka `g0/g1`: spectral bandwidth $\rho = +0.76 / +0.78$ (FNO), $+0.84 / +0.85$ (DNO+PF).
- Random sea (deep): max steepness $\rho = +0.63 / +0.84$.
- Stokes (finite): max steepness $\rho = +0.70 / +0.83$.
- BF `g0/g1`: peak wavenumber k_{peak} , $\rho \in [+0.33, +0.87]$.

Global correlations with kh or depth ($\rho \approx -0.4$) appear strong but are confounded — Tanaka has both shallow water and broad spectra. Within-regime, the load-bearing predictors are *structural* (bandwidth, steepness, peak wavenumber), not *distributional*. This is what high-steepness and broadband ICs need from the model’s nonlinear-in- η part — exactly the part of the current architectures that is under-parameterised relative to the polynomial structure of the true DNO.

2.3 Rollout error vs. initial wavenumber

A per-regime diagnostic picks three ICs per regime binned by initial mean wavenumber $\langle k \rangle(0) = \sum_k k |\hat{\eta}(k, 0)|^2 / \sum_k |\hat{\eta}(k, 0)|^2$ (low / mid / high) and compares two stacked panels over the rollout horizon: $\text{rel-}L^2$ η -error (log scale, top) and truth mean wavenumber $\langle k \rangle(t)$ (bottom). On Tanaka `g0` the wave’s mean wavenumber is essentially stationary in time, and the rollout error grows two orders of magnitude — more rapidly for higher- k ICs. The same pattern holds on the linear regime, where $\langle k \rangle(t)$ is exactly flat by construction, so the error growth is not riding on any spectral drift in the underlying wave. The full panel grid for all 10 regimes is on the companion site.

2.4 Data audit: not the bottleneck

A 1000-snapshot sample of the 73.7 GB training dataset found no NaN/inf in over 10^6 sampled voxels of η , ξ , or $\mathcal{G}\xi$. Every test regime’s IC distribution falls inside the training central 90% at the median for amplitude, steepness, and dominant wavenumber. Two minor tails: the maximum-amplitude Tanaka cases nip the training $p99$ in η amplitude; the finite-depth Stokes regime’s dominant wavenumber sits at the upper edge of training k (~ 26 , within the spectral cutoff of 64).

One eval-side issue surfaced: one of the 16 Benjamin–Feir g1 ICs has the *truth* integrator diverge to NaN at $t \approx 146$ of $T_{\max} = 200$.

Conclusion. The rollout-accuracy ceiling is not a data-coverage problem. Improvements have to come from the model or the training-time loss, not from acquiring more data.

2.5 Linear-regime truth fix

The legacy “linear” regime ICs are physically nonlinear shallow-water cases (η/h up to 0.30, $kh \approx 0.6$) inherited from a MATLAB test set. The Zakharov truth solver was diverging on some of them. The eval suite was updated to use analytic per-mode Fourier propagation of the linear Zakharov system,

$$\omega^2(k) = g k \tanh(hk), \quad \hat{\eta}(k, t) = \hat{\eta}_0(k) \cos \omega t + \frac{\mathcal{G}_{\text{lin}}(k)}{\omega} \hat{\xi}_0(k) \sin \omega t,$$

and analogously for $\hat{\xi}(t)$, as the linear-regime “truth”. Amplitude and top- k structure are preserved to better than 1% over the full horizon. All linear-regime comparisons in this document use the analytic truth.

2.6 Sensitivity sweep: is the FNO discrepancy a numerical artifact?

To rule out the obvious physical and numerical explanations for the rollout discrepancy, we ran a 3-axis sweep on Tanaka g0 case 0 (depth $h=0.139$, horizon $T=100$, output $dt=1$, FNO baseline). On each axis we hold the others fixed; the rel- L^2 η -error at $T = 100$ is:

Axis	Values				
IC amplitude α	0.50	0.75	1.00	1.50	2.00
rel- L^2 η @ T	0.0059	0.0050	0.0043	0.0080	0.0139
Integrator substeps / output dt	1	2	4	8	16
rel- L^2 η @ T	NaN	NaN	0.0044	0.0043	0.0042
Spatial N_x	512	1024	2048		
rel- L^2 η @ T	0.1435	0.0043	NaN @ $t=32$		

Read. The error is roughly α^2 on the large-amplitude side (consistent with nonlinear-in- η compounding) but stays within a factor of three across a $4\times$ amplitude range, with a minimum at the training amplitude $\alpha = 1$. Refining the integrator past substeps=4 does not change the error to four figures (the NaN at substeps $\in\{1, 2\}$ is the GL2 stepper itself being unstable at large dt, unrelated to the FNO). The FNO is, however, tightly resolution-locked to its training grid: at $N_x=512$ it is $33\times$ worse, and at $N_x=2048$ it blows up by $t=32$.

Conclusion. The discrepancy is invariant to numerical dt once the integrator is converged, and only modestly sensitive to physical amplitude. It is not a numerical artifact and is not amplitude-fragile; it is a model error. The N_x result is a separate finding — SpectralDNO/FNO are nominally resolution-invariant but in practice are tied to the training grid; this is worth flagging for any future multi-resolution evaluation.

3 Approach 1: One-step pushforward (training-loss)

3.1 Formulation

Given a data pair (η, ξ) , advance the surrogate’s prediction through k Euler steps of the Zakharov system, then re-predict $\mathcal{G}$ at the rolled-out state and penalise against the analytical Craig–Sulem expansion at order four:

$$\mathcal{L}_{\text{pf}} = \mathcal{L}\left(\mathcal{G}_{\theta}(\text{sg}(\eta_k), \text{sg}(\xi_k)), \mathcal{G}_{\text{CS}}(\eta_k, \xi_k)\right),$$

where (η_j, ξ_j) is the surrogate’s j -step Euler rollout. The total loss is $\mathcal{L}_0 + \lambda_{\text{pf}}\mathcal{L}_{\text{pf}}$. Gradients flow only through the final prediction at the rolled-out state; the trajectory (η_k, ξ_k) and the analytical target are both stop-gradient-wrapped.

3.2 Inductive bias

The single-step baseline only ever exposes the surrogate to data-distribution inputs. At rollout, the surrogate consumes its own predictions, where small per-step biases compound. The pushforward loss forces the surrogate to remain accurate at states it itself produces under iteration, breaking the covariate-shift compounding.

3.3 Implementation

One Euler step of the Zakharov system per pushforward step. The $\partial_t \eta$ term uses the surrogate’s own $\mathcal{G}$ -prediction; the $\partial_t \xi$ term uses the closed-form Zakharov RHS. The analytical target $\mathcal{G}_{\text{CS}}$ is the order-4 Craig–Sulem series with zero-padded FFTs at pad factor 4. The depth saturation $h_{\text{max}} = 5$ used inside the model is mirrored when building the analytical target so the target matches the operator the model is approximating.

3.4 Cost and tunables

$(1 + k)$ model forwards plus one vmapped order-4 Craig–Sulem evaluation per training step. Active configuration: $k = 1$, $\lambda_{\text{pf}} = 1$, $dt = 0.01$.

3.5 Status

A 40-epoch run from scratch is complete (best val 0.00490 at epoch 31). Per-regime rollout from that checkpoint: substantial improvement on Tanaka, Benjamin–Feir, modal, and the linear regime (median rel- L^2 reductions of 25–70%) but *regression* on random-sea regimes (near-zero error at FNO baseline grows to 0.09–0.12). The pushforward step appears to trade noise-robustness on broadband ICs for fidelity on coherent soliton-like ICs.

A 40-epoch resume to test for convergence was contaminated by a checkpoint-restore bug: the saved Adam moments and cosine-LR step counter did not deserialise cleanly, so the resume re-warmed Adam from zero and restarted the LR schedule from base lr. The restore path has since been fixed; an apples-to-apples 80-epoch run can now be done cleanly when GPU time frees up.

4 Approach 2: Hamiltonian-conservation loss (training-loss)

4.1 Formulation

The Zakharov system is Hamiltonian with

$$H(\eta, \xi) = \frac{1}{2} \int_{\Omega} (\xi \mathcal{G}(\eta) \xi + g \eta^2) dx.$$

An exact solution conserves H . The loss penalises relative drift of H over one Euler step of the surrogate’s own dynamics:

$$\mathcal{L}_H = \lambda_H w(s) \mathbb{E} \left[\min \left(\frac{|H_1 - H_0|}{|H_0| + \varepsilon}, C \right) \right],$$

with H_0 at the data state, H_1 at one Euler step (both $\mathcal{G}$ -values from the surrogate), linear warmup $w(s) = \min(s/s_{\text{warm}}, 1)$, and per-sample clip C . Total loss is $\mathcal{L}_0 + \mathcal{L}_H$.

4.2 Inductive bias

Even when pointwise $\mathcal{G}$ -error per step is small, accumulated rollout can violate energy conservation, producing non-physical growing or decaying waves. Penalising $|\Delta H|/|H|$ biases the surrogate toward predictions consistent with the symplectic structure across one step. This is a soft physical constraint, not a structural one — the architecture is unchanged.

4.3 Implementation

Two model forwards (at (η, ξ) and at (η_1, ξ_1)), two FFT pairs (to compute $\xi \mathcal{G}(\eta) \xi$ via the spectral inner product), and one closed-form Zakharov RHS evaluation. The advanced state is stop-gradient-wrapped; gradients flow only through the two $\mathcal{G}$ -predictions. The integrator and the depth saturation are treated as fixed.

4.4 Cost and tunables

Roughly $2\times$ the per-step cost of the single-step baseline. Active configuration: $\lambda_H = 1$, $dt = 0.01$, $s_{\text{warm}} = 10\,000$, $C = 10$.

4.5 Status

A first Modal training run reached epoch 10 (best val 0.00464, ahead of DNO+PF at the same epoch) before worker preemption. Partial evaluation on three regimes showed the rollout numbers tracking DNO+PF (random-sea regression of similar magnitude, slightly worse on Tanaka median, marginally higher divergence rate) — no breakthrough at that early checkpoint. A clean 25-epoch run from scratch is now training locally for a fair compute-matched comparison.

5 Approach 3: Craig–Sulem-shaped DNO architecture (model)

5.1 Formulation

The true Dirichlet–Neumann operator admits a Craig–Sulem expansion

$$\mathcal{G}(\eta; h)\xi = G_0(h)\xi + G_1(\eta, h)\xi + G_2(\eta \otimes \eta, h)\xi + \dots$$

where $G_0(h) = |D| \tanh(h|D|)$ is the flat-surface linear DNO and each G_n is multilinear in $\eta^{\otimes n}$ and linear in ξ , formed by alternating pointwise multiplications by powers of η with Fourier multipliers $M(D, h)$. The proposed architecture is a sum of learned Craig–Sulem-shaped blocks:

$$\mathcal{G}_\theta(\eta; h)\xi = G_0(h)\xi + \sum_{\ell=1}^{N_{\text{blocks}}} \sum_{i=1}^{n_{\text{branches}}} M_{\text{out}}^{\ell, i}(D, h) \left[\Phi^{\ell, i}(\eta)(x) \cdot (M_\xi^{\ell, i}(D, h)\xi)(x) \right],$$

where:

- $\Phi^{\ell, i}(\eta)$ is a learned pointwise projection from a shared η -feature stack (pointwise powers η, η^2, η^3 plus spectral derivatives $\partial_x \eta, \partial_x^2 \eta, |D|^{1/2} \eta, \mathcal{H}\eta$) to one spatial channel.
- $M_\xi^{\ell, i}$ and $M_{\text{out}}^{\ell, i}$ are learned depth-and- k -aware Fourier multipliers, each parameterised as a small MLP of the physical features $(k, h, \tanh(hk), k \tanh(hk))$ — the natural features of the linear dispersion relation, so the MLP only learns perturbations on the right ansatz.
- The $\Phi^{\ell, i} \rightarrow$ spatial channel projection is zero-initialised so each block starts inert; the model initially predicts the linear baseline exactly.

5.2 Inductive bias

Two simultaneous structural priors:

1. **Linear in ξ by construction.** The output residual is a sum of terms each of the form $(\eta\text{-only}) \cdot (M(D, h)\xi)$. This prevents the sideband hallucination observed in FNO on the linear regime.
2. **Polynomial in η by construction.** Each block expresses a learned G_n -shaped term; stacking blocks gives the architecture access to higher-order Craig–Sulem contributions — the very terms that dominate when steepness or bandwidth is large.

These together address the two strongest within-regime correlates of rollout failure identified in §2: bandwidth and steepness.

5.3 Implementation

The base SpectralDNO uses one rank-latent spatial $\times$ spectral sandwich per block; the new architecture replaces it with n_{branches} parallel *pointwise* CS-shaped terms per block. There is no tensor product between channel and branch axes — each branch carries one spatial channel and one ξ -multiplier, so parameter count grows linearly in n_{branches} . The η -feature stack is shared across blocks; per-block parameters are dominated by the two depth-aware multipliers and the Φ -projection.

5.4 Cost and tunables

A first run uses $\text{width} = 512$, $N_{\text{blocks}} = 8$, $n_{\text{branches}} = 256$, $\text{mult_hidden} = 128$ — about 1.1×10^6 parameters, an order of magnitude smaller than the FNO baseline (13.1M) and SpectralDNO (17.0M). The intent is to test whether the inductive bias is doing the heavy lifting; capacity can be scaled if the first signal is positive.

5.5 Status

Wired into the training and rollout-evaluation paths. Smoke checks confirm: $y(2\xi) = 2y(\xi)$ to numerical precision (linearity in ξ); $y(2\eta) - y(0) \neq 2(y(\eta) - y(0))$ after parameter perturbation (nonlinearity in η); $y = G_0\xi$ exactly at initialisation (zero-init blocks). 40-epoch training run in progress (epoch 11 at the time of writing, best val 0.00199 — already 46% below the FNO baseline’s all-time best of 0.00369, attained in 1/7th the training compute; training is in parallel with the H-cons retrain on a separate GPU).

6 Candidates for the next iteration

These are not currently running but are next in the queue based on the forensic findings and a literature scan.

6.1 Multi-step unrolled training with gradients through all steps

A direct extension of the one-step pushforward. Replace $k = 1$ stop-gradient pushforward with a k -step unroll where gradients flow through all steps, with a curriculum that ramps k from 1 to 8:

$$u \leftarrow u, \quad \mathcal{L} \leftarrow 0, \quad \text{for } j = 1 \dots k: \quad u \leftarrow \mathcal{G}_\theta(u), \quad \mathcal{L} += w_j \|u - u_j^{\text{true}}\|^2.$$

Stop-gradient one-step pushforward has a documented bias scaling like k^2 relative to the true multi-step gradient, and $k \geq 8$ is often needed for stable long rollouts. This is the most natural

single change to address the pushforward regression on random-sea regimes; memory scales linearly in k , but at the current model size $k = 4$ fits on a single H100.

6.2 Tendency-target reformulation

Reframe the surrogate’s output as a temporal derivative and integrate with a real RK4 stepper at inference. For the DNO setting, the natural place to apply this is on the ξ side: instead of using the closed-form Bernoulli RHS at rollout, have the surrogate also predict $\partial_t \xi$ and stitch with RK4 externally. This decouples the surrogate from carrying the dispersive phase rotation across one large step.

6.3 PDE-Refiner-style noise-conditioned refinement

Standard MSE training under-weights low-amplitude high- k modes — exactly the modes that drive modulational instability in Zakharov. Replace one-shot prediction with K iterative noise-conditioned denoising passes (same model called K times conditioning on a log-noise channel). Inference cost goes up by $K\times$. The published gain is reportedly smaller on FNO than U-Net, which is a concern for the spectral backbone here but less so given the Craig–Sulem architecture above.

6.4 Rejected directions

- Strict-symplectic / hard-Hamiltonian architectures — wrong constraint level for this discretisation; the soft H-conservation loss already covers it.
- Global Lipschitz / spectral-norm constraints — kills modulational dynamics for dispersive PDEs (true operator’s L^2 -induced norm is unitary).
- Simple capacity scaling — current SpectralDNO is 17M params vs. FNO’s 13M; capacity is not the bottleneck.

7 Comparison of approaches

Approach	Type	Constrains	Cost	Status
Pushforward (1-step)	loss	rollout state dist.	$\sim 2\times/\text{step}$	40 ep done; 80 ep gated on GPU
H-cons (1-step)	loss	energy drift	$\sim 2\times/\text{step}$	25-ep retrain running
Craig–Sulem DNO	model	poly- η , lin- ξ	$\sim 0.5\times$ base	40-ep run in progress (ep 11)
Multi-step unroll $k=4..8$	loss	rollout state dist.	$k\times/\text{step}$	queued
Tendency target + RK4	model	one-step ODE structure	$4\times$ at rollout	queued
PDE-Refiner	loss	high- k mode coverage	$K\times$ at rollout	queued

The three running approaches are not mutually exclusive. The two loss-side terms compose in the training script. Either can be paired with the Craig–Sulem architecture without changes — the model exposes the same $\mathcal{G}_\theta(\eta, \xi, h)$ signature that the loss terms rely on.

8 Open questions

- Does the Craig–Sulem architecture without any rollout loss already beat pushforward-trained SpectralDNO? If so, the polynomial structure is the operative inductive bias, not the training loss.
- Does the random-sea regression under pushforward survive the Craig–Sulem architecture, or is it an artifact of the SpectralDNO single-sandwich design?

- Should the H-cons functional use $\mathcal{G}_{\text{true}}\xi$ rather than $\mathcal{G}_{\theta}\xi$ when evaluating H_0 ? The current form rewards self-consistency of the surrogate’s energy belief, not conservation of the true energy.
- Are the Benjamin–Feir g1 / modal NaN blow-ups (FNO baseline) driven by the η update or the ξ update? A step-by-step trace would say which loss term has more leverage.

Companion artifacts

The per-regime rollout dashboard, including the error-vs-wavenumber plots, is hosted at stunning-cocada-db3a7d.netlify.app. Forensic reports (data audit, per-IC rollout breakdown, literature review) and JSON statistics behind every claim in §2 are available on request.